

Compressed Sensing

Consider a linear system

$$Ax = b$$

$$A \in \mathbb{R}^{n \times n}, \quad b \in \mathbb{R}^n$$

$$n > m$$

(More equations than unknowns).

WLOG: ~~$A \in \mathbb{R}^{n \times n}$~~ $\text{rank}(A) = m$

$\Downarrow$

A solution exists for any $b \in \mathbb{R}^m$.

But non-uniqueness is typical.

To select desirable solutions, look for sparse x .

Def $x \in \mathbb{R}^n$ is S -sparse if it has at most $S \leq n$ non-zero entries

The number of these is denoted by $\|x\|_0$.

The support of a vector is the set

$$\text{supp}(x) = \{j \in \{1, \dots, n\} : x_j \neq 0\}$$

(!!) $\|\cdot\|_0$ is not a norm.

Cardinality $|\text{supp}(x)| := \|x\|_0$

Next, given a set $S = \{j_1, \dots, j_k\}$, denote

x_S is either the vector in $\mathbb{R}^S$ containing the non-zero components of x , or

its extension to a vector in $\mathbb{R}^n$ with 0 entries for indices not from S .

Columns of $A \in \mathbb{R}^{m \times n}$ are denoted
by $a_j \in \mathbb{R}^m$ ($1 \leq j \leq n$)

$A_S \in \mathbb{R}^{m \times s}$ the matrix consisting of
columns of A with index in S

$$A_S = (a_{j_1}, a_{j_2}, \dots, a_{j_s}) \in \mathbb{R}^{m \times s}$$

Def A vector $x^* \in \mathbb{R}^n$ is an sparse
solution of

$$Ax = b$$

if it solves

$$\min \{ \|x\|_0 : x \in \mathbb{R}^n, Ax = b \}.$$

(!?) This problem is NP-hard

Def The problem of basis pursuit is to find a solution x^* of

$$(*) \quad \min \{ \|x\|_1 : x \in \mathbb{R}^n, Ax = b \}.$$

Lemma Let $A \in \mathbb{R}^{m \times n}$, $n > m = \text{rank}(A)$.

Then $(*)$ has a solution for every $b \in \mathbb{R}^m$.

Pf Since $\text{rank}(A) = m$, $\exists x_0 \in \mathbb{R}^n$ such that $Ax_0 = b \Rightarrow$

The set $K := \{x \in \mathbb{R}^n; Ax = b, \|x\|_1 \leq \|x_0\|_1\}$

is non-empty.

K is closed and bounded $\Rightarrow$

K is compact.

The continuous function $x \mapsto \|x\|_1$

attains its minimum ~~on~~ⁱⁿ K . □

Next, consider $A \in \mathbb{R}^{m \times n}$, $n > m$, $b \in \mathbb{R}^m$.

Thm Assume that $(*)$ has a unique solution $x^* \in \mathbb{R}^n$.

Then, the columns a_j of A with $j \in \text{supp}(x^*)$ are linearly independent. In particular, x_* is sparse with $\|x^*\|_0 \leq m$.

Pf. (Contradiction). Assume that a_j with $j \in S := \text{supp}(x^*)$ are linearly dependent.

Then $\exists v \in \mathbb{R}^n \setminus \{0\} : \text{supp}(v) \subseteq S$

and $Av = 0$.

Thus $x_t = x_* + v$ satisfies $Ax_t = b$

$\forall t \in \mathbb{R}$.

Therefore, uniqueness

$$\|x^*\|_1 < \|x_t\|_1 = \|x^* + tv\|_1 = \sum_{j \in S} |x_j^* + tv_j| = \sum_{j \in S} \text{sgn}(x_j^* + tv_j) (x_j^* + tv_j)$$

Since all x_j^* are $\neq 0$, we can choose $t \neq 0$ with $|t|$ so small that $\text{sgn}(x_j^* + tv_j) = \text{sgn}(x_j^*)$ $\forall j \in S$.

Then

$$\begin{aligned} \|x^*\|_1 &< \sum_{j \in S} \text{sgn}(x_j^*) x_j^* + t \sum_{j \in S} \text{sgn}(x_j^*) v_j \\ &= \|x^*\|_1 + t \sum_{j \in S} \text{sgn}(x_j^*) v_j \end{aligned}$$

Now we can choose t so that the second term in RHS is $<$

Thus

$$\cancel{\|x^*\|_1} \cdot \|x^*\|_1 < \|x^*\|_1,$$

a contradiction

□.

Next, Suppose that $b = Ax^*$ where x^* is S -sparse.

Is this x^* a unique solution of (*)?

Yes, if A satisfies a certain condition.

Def A matrix $A \in \mathbb{R}^{m \times n}$ ($m > n$) and satisfies

a null space property w.r. to a set $S \subseteq \{1, \dots, n\}$ if

$$\|v_s\|_1 \leq \|v_{sc}\|_1$$

$$\forall v \in \ker(A) \setminus \{0\}.$$

The matrix A satisfies the null space property of order s if it satisfies the N.S.P. for each $S \subseteq \{1, \dots, n\}$ with $|S| \leq s$.

Note: $\|v_{sc}\|_1 + \|v_s\|_1 = \|v\|_1,$

Thus

$$\|v_s\|_1 < \|v_{sc}\|_1$$



$$2\|v_s\|_1 < \|v_{sc}\|_1 + \|v_s\|_1 = \|v\|_1$$



$$\|v_s\|_1 < \frac{1}{2} \|v\|_1.$$

$$\forall v \in \ker(A) \setminus \{0\}.$$

(Thm) Suppose that

$A \in \mathbb{R}^{m \times n}$, $n > m$, and $S \subseteq \{1, \dots, n\}$

Then the following statements are equivalent

1. $\forall x \in \mathbb{R}^n$ with $\text{supp}(x) \subseteq S$, the problem (*) with $b := Ax$ has the unique solution x
2. The matrix A satisfies the N.S.P w.r. to S .

pf ($\Rightarrow$) Assume 1. Then $\forall x \in \text{Ker}(A) \setminus \{0\}$

$x_S \in \mathbb{R}^n$ is the unique solution of

$$\min \{ \|z\|_1 \text{ subject to } Az = Ax_S \}$$

$$0 = Ax = Ax_S + Ax_{sc}$$

implies

$-x_{sc}$ satisfies

$$A(-x_{sc}) = Ax_S$$

Since x_S is the unique solution of the minimization problem, we conclude that

$$\|x_S\|_1 < \|-x_{sc}\|_1 = \|x_{sc}\|_1$$

(this is N.S.P. 2.)

2. Assume 2. let $x \in \mathbb{R}^n$ with $\text{supp}(x) \subseteq S$ be given.

Assume that $\nexists z \in \mathbb{R}^n$, $z \neq x$,

$$Az = Ax$$

Then $v = x - z \in \text{Ker}(A) \setminus \{0\}$.

Since $x_{SC} = 0$ and using N.S.P. we have

$$\begin{aligned} \|x\|_1 &= \|x - z_g\|_1 + \|z_g\|_1 = \\ &= \|v_g\|_1 + \|z_g\|_1 \end{aligned}$$

$$\begin{aligned} & \textcircled{<} \|v_{sc}\|_1 + \|z_s\|_1 = \|z_{sc}\|_1 + \|z_s\|_1 \\ & = \|z\|_1 \end{aligned}$$

$\rightarrow x$ is the unique minimizer. $\square$

Corollary Every s -sparse vector $x \in \mathbb{R}^n$ is the unique solution of $(*)$ with $b := Ax$ iff.

A satisfies the N.S.P. of order s .

Thm Let $A \in \mathbb{R}^{m \times n}$, $n > m$
 satisfy the NSP of order s .

Let $x \in \mathbb{R}^n$ be an s -sparse
 vector and let $b := Ax$.

Then, the solution of

$$\min \{ \|z\|_0 : z \in \mathbb{R}^n, Az = b \}$$

is given by x .

Moreover, x is the solution of
 the basis pursuit problem

$$\min \{ \|z\|_1 : z \in \mathbb{R}^n, Az = b \}.$$

Pf. Assume that $\exists z \in \mathbb{R}^n$ with

$$Az = b = Ax \text{ and } \|z\|_0 \leq \|x\|_0$$

Then z is also s -sparse, and
 by the previous corollary,

$$z = x.$$

□

The situation so far:
 if RHS $b = Ax$ for x ^{that is} s -sparse,
 then and A satisfies NSP of order
 S , then the unique solution of
 Basis Pursuit

$$\min \{ \|z\|_1 : Az = b \}$$

satisfies also solves

$$\min \{ \|z\|_0 : Az = b \}.$$

Q: What can we do if b is
 not so generalized?

Def Let $x \in \mathbb{R}^n$ and $1 \leq s \leq n$. Then
 the best approximation error of
 x from the set of all s -sparse
 vectors is denoted by

$$\sigma_s(x) := \min \{ \|x - z\|_2 : z \in \mathbb{R}^n, z \text{ sparse} \}$$

We call $x \in \mathbb{R}^n$ compressible if $\sigma_s(x)$ decays rapidly for $s \uparrow$.

(!!) Minimum in the ~~of~~ of $\sigma_s(x)$ is attained by z containing exactly s largest absolute values of components of x .

Next, we strengthen the NSP requirement as follows.

Def A matrix $A_{m \times n}$ with $n > m$ satisfies the stable null space property with constant $0 < \rho < 1$ w.r. to a set $S \subseteq \{1, \dots, n\}$ if

$$\|v_s\|_1 \leq \rho \|v_{s^c}\|_1$$

$\forall v \in \ker(A).$

- A satisfies the SNSP of order s with constant $0 < \rho < 1$ if it satisfies SNSP with ρ w.r. to all $S \subseteq \{1, \dots, n\}$ with $|S| \leq s$.

(Thm) let $A_{m \times n}$ with $n > m$ satisfy the SNSP with $\rho \in (0, 1)$ of order s .
Let $x \in \mathbb{R}^n$ and $b := Ax \in \mathbb{R}^m$.

Then the solution x^* of the basis pursuit problem satisfies

$$\|x - x^*\|_1 \leq 2 \frac{1+\rho}{1-\rho} \sigma_s(x).$$

Pf let $z \in \mathbb{R}^n$ be a vector
satisfying $Az = b = Ax$, $z \neq x$.

Then $v := z - x \in \ker(A)$. $\Rightarrow$

$$\|v_s\|_1 \leq \rho \|v_{s^c}\|_1$$

$\forall S \subseteq \{1, \dots, n\}, |S| \leq 8$.

Next, observe that

$$(1) \quad \|v_{s^c}\|_1 \leq \|x_{s^c}\|_1 + \|z_{s^c}\|_1$$

Δ ineq.

and

$$(2) \quad \|x\|_1 \leq \|x_{s^c}\|_1 + \|x_S\|_1 \leq \|x_{s^c}\|_1 + \|v_s\|_1 + \|z_S\|_1$$

Δ as well

Add (1) and (2)

$$\|x\|_1 + \|v_{s^c}\|_1 \leq 2\|x_{s^c}\|_1 + \|v_s\|_1 + \|z\|_1$$

Now use SNSP to get

$$\|V_{sc}\|_2 \leq \|z\|_1 - \|x\|_1 + \|V_s\|_2 + 2\|x_{sc}\|_1$$

$$\stackrel{\text{SNSP}}{\leq} \|z\|_1 - \|x\|_1 + \rho \|V_{sc}\|_1 + 2\|x_{sc}\|_1$$

~~Rearrange:~~ $\hookrightarrow$

$$\|V_{sc}\|_1 \leq \frac{1}{1-\rho} (\|z\|_1 - \|x\|_1 + 2\|x_{sc}\|_1)$$

$$(1-\rho)\|V_{sc}\|_1 \leq$$

$$\leq \|z\|_1 - \|x\|_1 + 2\|x_{sc}\|_1.$$

Then

~~SNVP~~
~~ASAP~~

$$\|v\|_1 = \|v_S\|_1 + \|v_{S^c}\|_1 \leq$$

$$\leq (1+p) \|v_{S^c}\|_1 \leq \frac{1+p}{1-p} (\|z\|_1 - \|x\|_1 + 2\|x_{S^c}\|_1).$$

Now pick S to be the set of indices corresponding to the s largest entries of x , so that

$$\|x_{S^c}\|_1 = \sigma_s(x),$$

and let $z = x^*$ be the solution of the basis pursuit problem

Then $Ax^* = Ax$ and $\|x^*\|_1 \leq \|x\|_1$

($\Rightarrow \|z\|_1 - \|x\|_1 \leq 0$). Thus

$$\|v\|_1 \leq 2 \frac{1+p}{1-p} \|x_{S^c}\|_1.$$

□

Next, relax the basis pursuit problem.

Def let $A_{m \times n}$ with $n \geq m$ and $b \in \mathbb{R}^m$ be given. For a given $\gamma > 0$, the robust basis pursuit problem is

$$(3) \quad \min \|z\|_1 : z \in \mathbb{R}^n \text{ with } \|Az - b\| \leq \gamma$$

(the norm ⁱⁿ $\|Az - b\|$ is not necessarily 1-norm).

• if the set $\{z \in \mathbb{R}^n : \|Az - b\| \leq \gamma\}$ is non-empty (this will be ^{in particular} true if $\text{rank}(A) = m$), then the problem

(3) has a solution

Def $A_{m \times n}$ with $n > m$ satisfies

the robust null space property

with constants $0 < \rho < 1$ and

$\tau > 0$ w.r.t. a set $S \subseteq \{1, \dots, n\}$
if

$$(4) \quad \|v_S\|_1 \leq \rho \|v_{S^c}\|_1 + \tau \|Av\|, \quad v \in \mathbb{R}^n$$

As before, RNSP of order S

is defined as above.

- if $v \in \ker(A)$, then (4) reduces to SNSP.

Thm let $A_{m \times n}$ with $n > m$ satisfy the RNSP of order S with constants $0 < \rho < 1$ and $\tau > 0$

For $x \in \mathbb{R}^n$ let $b = Ax + e$

with $\|e\| \leq \eta$.

Then, any solution x^* of the robust basis pursuit problem satisfies the error bound

$$\|x - x^*\|_1 \leq 2 \frac{1+p}{1-p} \sigma_s(x) + \frac{4\tau}{1-p} \eta.$$

Pf Starting with

$$\begin{aligned} \|v_{sc}\|_1 &\leq \|z\|_1 - \|x\|_1 + \|v_s\|_1 + 2\|x_{sc}\|_1 \\ &\leq \|z\|_1 - \|x\|_1 + 2\|x_{sc}\|_1 + \\ &\quad + p\|v_{sc}\|_1 + \tau\|Av\| \end{aligned}$$

$\Downarrow$

$$(1-p)\|v_{sc}\|_1 \leq \|z\|_1 - \|x\|_1 + 2\|x_{sc}\|_1 + \tau\|Av\|$$

Thus

$$\|v\|_1 = \|V_S\|_1 + \|V_{S^c}\|_1 \leq$$

$$\stackrel{RNSP}{\leq} (1+\rho) \|V_{S^c}\|_1 + \tau \|Av\| \leq$$

$$\leq \frac{1+\rho}{1-\rho} \left(\|z\|_1 \|x\|_1 + 2 \|x_{S^c}\|_1 \right) + \frac{2\tau}{1-\rho} \|Av\|$$

Choose S now again as in the pf of the previous thm so that

$$\|x_{S^c}\|_1 = \sigma_S(x)$$

and let $z = x^*$ be a solution of the robust pursuit problem.

Then

$$\|Av\| = \|Ax^* - Ax\| \leq \|Ax^* - b\| + \|b - Ax\| \leq$$

$$\leq 2\eta$$

□

Restricted Isometry

Isometry: a linear mapping

$A: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is isometric

if $\|Ax\| = \|x\|$.

(~~for any~~ relative to a specific choice of norms).

Def A matrix $A_{m \times n}$ satisfies the

restricted isometry property of order

s if there is $\delta \in (0, 1)$ such that

$$(**) \quad (1-\delta)\|x\|_2^2 \leq \|Ax\|_2^2 \leq (1+\delta)\|x\|_2^2$$

is satisfied for all s -sparse

vectors $x \in \mathbb{R}^n$.

The smallest $\delta \in (0, 1)$ in $(**)$ is denoted by δ_s (restricted isometry constant)

(!!) A matrix satisfying

(*) of order $2s$ allows us

to uniquely recover an s -sparse vector.

Lemma Suppose that $s \in \mathbb{N}$ and that $A_{m \times n}$ ($n \geq m$) satisfies (**) of order $2s$ with $\bar{\sigma}_{2s} < 1$.

If $x \in \mathbb{R}^n$ is s -sparse, then

x is uniquely determined by

$$b := Ax.$$

Pf Let $S := \text{supp}(x)$. Then $|S| \leq s$

and we need to show that

there is no other $y \in \mathbb{R}^n$, $\|y\|_0 \leq s$

and $Ay = b$.

By contradiction.

Assume that such y exists,
with $\tilde{S} := \text{supp}(y)$, $|\tilde{S}| \leq S$.

Then $z = x - y \in \mathbb{R}^n$ has

$$|\text{supp}(z)| \leq 2S \quad (\text{supp}(z) \subseteq S \cup \tilde{S})$$

and $Az = 0$. Then

from $(*)$,

$$(1 - \delta_{2S}) \|z\|_2^2 \leq \|Az\|_2^2 = 0$$

$\Downarrow$

$$z = 0.$$

□.

Next, observe that ℓ_0 -minimization
is NP-hard, and the estimate

$$\lambda_{\min}(A^T A) \|x\|_2^2 \leq \|Ax\|_2^2 \leq x^T A^T A x \leq \lambda_{\max}(A^T A) \|x\|_2^2$$

is not helpful in finding δ_S .

Lemma if $A_{n \times n}$ satisfies $(**)$ of order S then the restricted isometry constant S is given by

$$\delta_S = \max \{ \|A_S^T A_S - I\|_2 : S \subseteq \{1, \dots, n\}, |S| \leq s \}.$$

pf. Let $S \subseteq \{1, \dots, n\}$ with $|S| \leq s$ be given. Since $A_S^T A_S - I$ is symmetric we can write

$$\begin{aligned} \|A_S^T A_S - I\|_2^2 &= \max_{\|x\|_2=1} | \langle (A_S^T A_S - I)x, x \rangle | \\ &= \max_{\|x\|_2=1} | \|A_S x\|_2^2 - \|x\|_2^2 | \end{aligned}$$

Thus, A satisfies $(**)$ of order S iff.

$$\|A_S^T A_S - I\|_2 < 1$$

for all $S \subseteq \{1, \dots, n\}$
with $|S| \leq s$.

Thus δ_s is given by

$$\delta_s = \max \{ \|A_S^T A_S - I\|_2 : S \subseteq \{1, \dots, n\}, |S| \leq s \}$$

□.

Proposition (needed for the following
thm). Let $A_{n \times n}$ be given. Let

$$x, y \in \mathbb{R}^n, \|x\|_0 \leq s, \|y\|_0 \leq t.$$

If the supports of ~~x and y~~ and x ^{and} y are disjoint, then

$$|\langle Ax, Ay \rangle_2| \leq \delta_{s+t} \|x\|_2 \|y\|_2$$

pf let $S = \text{supp}(x) \cup \text{supp}(y)$.

then $|S| = s + t$

By disjoint support, $\langle x, y \rangle = \langle x_S, y_S \rangle =$

Thus

$$|\langle Ax, Ay \rangle_2| = |\langle A_S x_S, A_S y_S \rangle_2 - \langle x_S, y_S \rangle_2|$$

$$= |\langle (A_S^T A_S - I) x_S, y_S \rangle_2| \leq$$

$$\leq \|A_S^T A_S - I\|_2 \|x_S\|_2 \|y_S\|_2.$$

The result follows from the previous Lemma

□

(Thm) Let $A_{m \times n}$ be given.

Suppose that its restricted isometry constant δ_{2s} of order $2s$ satisfies $\delta_{2s} < \frac{1}{3}$.

Then A satisfies the NSP of order s . Therefore, any s -sparse $x \in \mathbb{R}^n$ is the unique solution of the basis pursuit problem with RHS $b := Ax$.

pf. ~~We~~ It is enough to show

$$\|v_s\|_2 < \frac{1}{2} \|v\|_2$$

(which is equivalent to $\|v_s\|_2 < \|v_{s^c}\|_2$
 $\forall v \in \ker(A) \setminus \{0\}$).

If $x \in \ker(A) \setminus \{0\}$, it is enough to prove this for $S = S_0$ which contains the indices of the s largest ~~the~~ abs. entries of x .

Next, let S_1 denote the set of indices of the s largest abs. entries of S^c . Then, let S_2 denote the indices of s largest abs. entries of $(S_0 \cup S_1)^c$ and so on.

Since $x \in \ker(A)$, we have

$$Ax_S = A(-x_{S_1} - x_{S_2} - \dots) = -\sum_{k \geq 1} Ax_{S_k}$$

Viewing x_{S_0} as a $2s$ -sparse vector, this, together with

$$(1-\delta) \|x\|_2^2 \leq \|Ax\|_2^2 \leq (1+\delta) \|x\|_2^2$$

and the previous Prop.,

$$\begin{aligned}
 \|X_{S_0}\|_2^2 &\leq \frac{1}{1-\delta_{2S}} \|AX_{S_0}\|_2^2 = \\
 &= \frac{1}{1-\delta_{2S}} \langle AX_{S_0} - \sum_{k \geq 1} AX_{S_k} \rangle \\
 &\leq \frac{1}{1-\delta_{2S}} \sum_{k \geq 1} |\langle AX_{S_0}, AX_{S_k} \rangle_2| \\
 &\leq \frac{\delta_{2S}}{1-\delta_{2S}} \|X_{S_0}\|_2 \sum_{k \geq 1} \|X_{S_k}\|_2
 \end{aligned}$$

i.e.

$$\|X_{S_0}\|_2 \leq \frac{\delta_{2S}}{1-\delta_{2S}} \sum_{k \geq 1} \|X_{S_k}\|_2 \quad (+)$$

Now, the specific sorting of the index sets S_k allows us to bound

$$\|X_{S_k}\|_2 = \left(\sum_{j \in S_k} |x_j|^2 \right)^{1/2}$$

$$\leq \sqrt{s} \max_{j \in S_k} |x_j| \leq \sqrt{s} \max_{j \in S_{k-1}} |x_j|$$

$$\leq \sqrt{s} \frac{1}{s} \sum_{j \in S_{k-1}} |x_j| = \frac{1}{\sqrt{s}} \|X_{S_{k-1}}\|_1.$$

Inserting this into (+) we obtain

$$\|X_{S_0}\|_2 \leq \frac{\delta_{2s}}{1-\delta_{2s}} \frac{1}{\sqrt{s}} \sum_{k \geq 1} \|X_{S_{k-1}}\|_1 =$$

$$= \frac{\delta_{2s}}{1-\delta_{2s}} \frac{1}{\sqrt{s}} \|X\|_1$$

and hence

$$\|X_{S_0}\|_1 \leq \sqrt{s} \|X_{S_0}\|_2 \leq \frac{\delta_{2s}}{1-\delta_{2s}} \|X\|_1.$$

The result now follows from the fact that

$$\delta_{2s} \frac{1}{1-\delta_{2s}} < \frac{1}{2} \Leftrightarrow \delta_{2s} < \frac{1}{3}$$